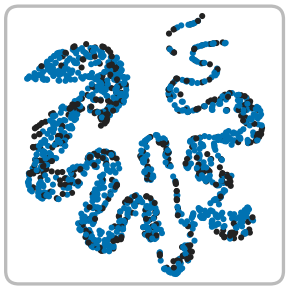
Outputs per cell and per gene, from one fit

per cell: embedding z



by species
- human
- monkey



by cell type
- hepatocytes
- endothelial
- fibroblasts
- macrophages
- other

species aligned: clustering, label transfer; the substrate of the gene readouts

per gene: five example orthologs (schematic)

count evidence S_g  none

prior π_g: tick
gate α_g: needle

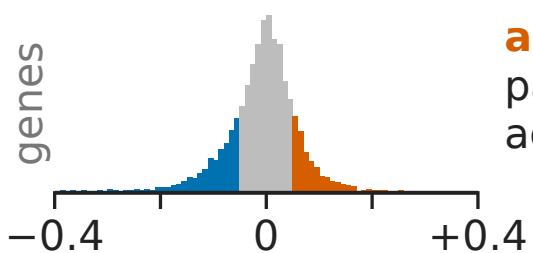

α_g from 0 (full residual) to 1 (shared decoder only)

$\Delta\alpha_g = \alpha_g - \pi_g$

protein-feature gate	+0.30	+0.13	−0.16	−0.34	*+0.13*
per-gene readout	+0.28	+0.09	−0.12	−0.31	0

gray: imputed from protein features

below the prior
pattern or level parts between species



above the prior
pattern shared across species

$\Delta\alpha_g$ of one model: liver, human + monkey (8,430 genes)